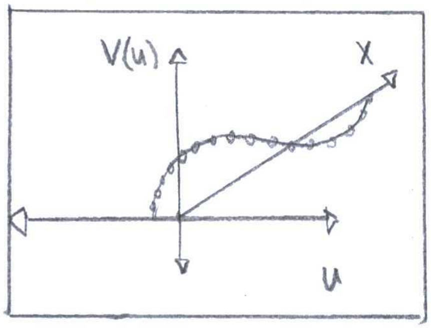
Chapter 3: Path Integral:

Exercise #1: (Pg 101)



"Path integral"

Position: $y(X,t) = A\sin(kX - \omega t)$

Velocity: $\dot{y}(X,t) = \omega A\cos(kX - \omega t)$

Kinetic Energy:

$$K.E. = \int \frac{1}{2}\mu\left(-\omega A\cos(kX-\omega t)\right)^2 dX$$

... when $\mu = \frac{\text{mass}}{\text{Length}}$

$$= \frac{1}{2}\mu\omega^2A^2\int_0^L \cos^2(kX-\omega t)\,dX$$

$$= \frac{1}{2}\mu\omega^2A^2\left(\frac{1}{2}L + \frac{\sin(2kL)}{4k}\right)$$

$$\boxed{\text{Wavevector: } k = 2\pi/L}$$

$$= \frac{1}{4}\mu\omega^2A^2$$

The continuum lattice model in Chapter 1 had an integral over a "path" from 0 to L, but with a Hamitonian rather than a direct integral to Kinetic energy.

Limit ($m \to 0$):

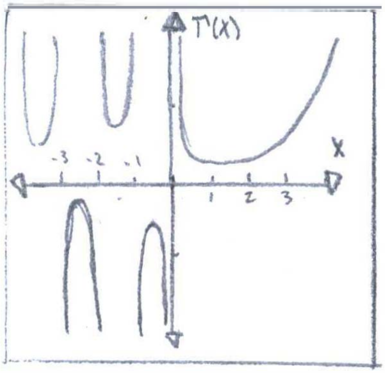
$$\lim_{m\to\infty} Z = \lim_{m\to\infty} tr\left(e^{-\beta H}\right)$$

$$= \lim_{m\to\infty} tr\left(e^{-\beta(KE+V)}\right)$$

$$= \lim_{m\to\infty} tr\left(e^{-\beta\left(\frac{1}{4}\mu\omega^2 A^2 + \hat{V}\right)}\right)$$

$$= 0$$

Exercise #2:
(pg 105)



"Gamma Function"

(from problem) "Gamma Function"-modified"

$$\Gamma(z+1) = \int_0^\infty dx \cdot x^z e^{-x}$$

Sterling Approximation:

$$\Gamma(z+1) = \int_0^\infty dx \cdot x^z \cdot e^{-x}$$

$$= \int_0^\infty dx \cdot e^{-x + s\ln x}$$

$$\cong \int_0^\infty dx \cdot e^{-s + s\ln s - x^2/2s}$$

$$\cong e^{s(\ln s - 1)} \cdot \sqrt{2\pi s}$$

Minimum to Exponent:

$$F(x) = -x + s\ln x$$

$$F'(x) = -1 + \frac{s}{x}$$

$$= 0$$

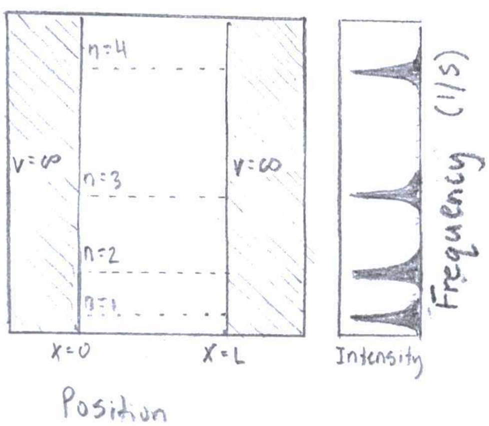
$$x_{min} = s$$

Integral of a Gaussian Distribution

$$\int e^{-ax^2} dx = \sqrt{\frac{\pi}{a}}$$

Exercise #3

(pg 107)



"Quantum particle in a well"

(3.27) "Free Particle Operator"

$$G_{free}(q_F ; q_i , t) = \langle q_F | e^{-\frac{i}{\hbar}\frac{\hat{p}^2 t}{2m}} | q_i \rangle \Theta(t)$$

$$= \left(\frac{m}{2\pi i \hbar t}\right)^{1/2} e^{\frac{i}{\hbar}\frac{m(q_F-q_i)^2}{2t}} \circ \Theta(t)$$

"This integral"

Heavyside Θ-function

$$\langle q_F | e^{-\frac{i}{\hbar}\frac{\hat{p}^2 t}{2m}} | q_i \rangle = \int_{-\infty}^{\infty} \frac{e^{-\frac{i q_F p}{\hbar}}}{\sqrt{2\pi}} \circ e^{-\frac{i}{\hbar}\frac{p^2 t}{2m}} \circ \frac{e^{\frac{i q_i p}{\hbar}}}{\sqrt{2\pi}} \, dp$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\frac{i}{\hbar}\left[\frac{p^2 t}{2m} + \frac{(q_F-q_i)p}{} \left(\frac{t}{t}\right)\right]} dp$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\frac{i}{\hbar}\left[\frac{1}{2m}\left(p - \frac{m(q_F-q_i)}{t}\right)^2 - \frac{m(q_F-q_i)^2}{2t^2}\right] t} dp$$

$$= \frac{e^{-\frac{i}{\hbar}\frac{m(q_F-q_i)^2}{t}}}{2\pi} \int_{-\infty}^{\infty} e^{-\frac{i}{\hbar}\left[\frac{1}{2m}\left(p - \frac{m(q_F-q_i)}{t}\right)^2\right]} dp$$

$$= \frac{e^{\frac{i}{\hbar}\frac{m(q_F-q_i)^2}{t}}}{2\pi} \int_{-\infty}^{\infty} e^{\frac{-i p'^2 t}{2\hbar m}} dp'$$

$$\circ\circ\circ \; p' = p - \frac{m(q_F-q_i)^2}{t}$$

$$= \frac{e^{\frac{i}{\hbar}\frac{m(q_F-q_i)^2}{2t}}}{2\pi}\left(\frac{2\pi}{it/\hbar m}\right)$$

$$\boxed{\begin{array}{l}(3.9)\text{ "Gaussian Integral"} \\ \int_{-\infty}^{\infty} dx\, e^{-\frac{a}{2}x^2} = \sqrt{\frac{2\pi}{a}}\end{array}}$$

$$= \left(\frac{m}{2\pi i\hbar t}\right)^{1/2} e^{\frac{i}{\hbar}\frac{m(q_F-q_i)^2}{2t}}$$

$$G_{Free}(q_F, q_i, t)\,[\text{units} = \hbar^{-1}] = \left(\frac{m}{2\pi i\hbar t}\right)^{1/2} e^{\frac{i}{\hbar}\frac{m(q_F-q_i)^2}{2t}}$$

Exercise #4:
(pg 107)

(3.1) "Time evolution operator"

$$U(t, t') = e^{-i/\hbar \cdot \hat{H}(t-t')} \cdot \Theta(t'-t)$$

(3.7) "Action Integral"

$$\langle q_F | e^{-i\hat{H}t/\hbar} | q_i \rangle = \int_{\substack{q(t)=q_F \\ q(0)=q_i}} e^{-\frac{i}{\hbar}\int_0^t dt'\, V(q)} \cdot dq$$

$$\langle p' | U(t,t') | p \rangle = \langle p' | e^{-\frac{i}{\hbar}\hat{H}t} | p \rangle$$

$$= \langle p' | e^{-\frac{i}{\hbar}\int_0^t V(p)\,dt} | p \rangle$$

$$= \langle p' | 1 - \frac{i}{\hbar}\int_{t_0}^{t} V(p)\,dt | p \rangle$$

$$\boxed{\begin{array}{l}\underline{\text{Taylor Expansion:}} \\ e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots\end{array}}$$

$$= \langle p' | | p \rangle - \frac{i}{\hbar} \langle p' | \int_{t_0}^{t} V(p)\, dt \, | p \rangle$$

$$= -\frac{i}{\hbar} \langle p' | \int_{-\infty}^{\infty} e^{-iH_0 t/\hbar} \cdot V \cdot e^{iH_0 t/\hbar} | p \rangle$$

$\boxed{\begin{array}{l} \text{(Orthoganal)} \\ \underline{\text{Inner products}}: \\ \langle p' | | p \rangle = \delta(p'-p) \\ \quad = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases} \end{array}}$

$$= -\frac{i}{\hbar} \int_{-\infty}^{\infty} e^{-iE(p)t/\hbar} \cdot \langle p' | V | p \rangle \cdot e^{iE(p)t/\hbar} \, dt$$

$$= -\frac{i}{\hbar} \langle p' | V | p \rangle \int_{-\infty}^{\infty} e^{i(E(p') - E(p))t/\hbar} \cdot dt$$

$$\lim_{t \to \infty} \langle p' | U(t,t') | p \rangle = \lim_{t \to \infty} -\frac{i}{\hbar} \langle p' | V | p \rangle \int_{-\infty}^{\infty} e^{i(E(p')-E(p))t/\hbar} \cdot dt$$

$$\propto \delta(E(p) - E(p')) \quad \text{... say zero}$$

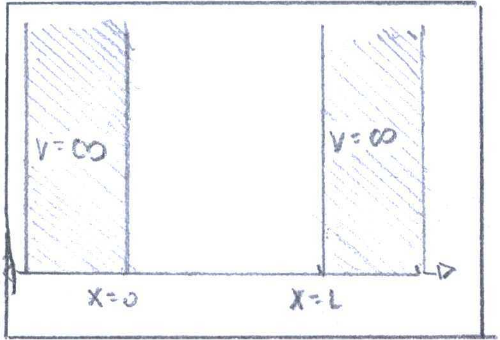
$$\lim_{t' \to -\infty} \langle p' | U(t,t') | p \rangle = \lim_{t \to \infty} -\frac{i}{\hbar} \langle p' | V | p \rangle \int_{-\infty}^{\infty} e^{i(E(p')-E(p))t/\hbar} \, dt$$

$$\propto \delta(E(p) - E(p')) \quad \text{... say one.}$$

(From problem) "Born scattering amplitude

$$\langle p' | U(t \to \infty, t' \to -\infty) | p \rangle = -\frac{i}{\hbar} e^{-i(t-t')E(p)/\hbar} \, \delta(E(p) - E(p')) \langle p' | V | p \rangle$$

Exercise #5:
(pg 109)



"infinite square well"

(3.27) "Free Particle Propagator"

$$G_{free}(q_F; q_i; t) = \langle q_F | e^{-\frac{i}{\hbar}\frac{\hat{p}^2 t}{2m}} | q_i \rangle \Theta(t)$$

$$= \int_{-\infty}^{\infty} \frac{e^{-\frac{i}{\hbar} p q_F}}{\sqrt{2\pi\hbar}} \cdot e^{-\frac{i}{\hbar}\frac{\hat{p}^2 t}{2m}} \cdot \frac{e^{\frac{i}{\hbar} p q_i}}{\sqrt{2\pi\hbar}} \, dp$$

$$= \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} e^{-\frac{i}{\hbar}\frac{\hat{p}^2 t}{2m}} \cdot e^{-\frac{i}{\hbar} p (q_F - q_i)} \, dp$$

As an Ensemble,

$$\sum_{m=-\infty}^{\infty} \left[\langle q_F | e^{-\frac{i}{\hbar}\frac{\hat{p}^2 t}{2m}} | q_i \rangle - \langle q_i | e^{-\frac{i}{\hbar}\frac{\hat{p}^2 t}{2m}} | q_F \rangle \right] e^{-\frac{2ipmL}{\hbar}} \, dp$$

$$= \frac{\sum_{m=-\infty}^{\infty}}{2\pi\hbar} \int_{-\infty}^{\infty} e^{-\frac{i}{\hbar}\frac{\hat{p}^2}{2m} t} \left[e^{\frac{i}{\hbar} p q_F} - e^{-\frac{i}{\hbar} p q_F} \right] e^{-\frac{i}{\hbar} p q_i} \cdot e^{-\frac{2ipmL}{\hbar}} \, dp$$

$$= \frac{i}{\pi\hbar} \int_{-\infty}^{\infty} e^{-\frac{i}{\hbar}\frac{\hat{p}^2}{2m} t} \cdot \sin\left(\frac{i}{\hbar} p \cdot q_F\right) \cdot e^{-\frac{i}{\hbar} p q_i} \sum_{m=-\infty}^{\infty} \cdot e^{-\frac{2ipmL}{\hbar}} \, dp$$

$$= \frac{i}{\pi\hbar} \int_{-\infty}^{\infty} e^{-\frac{i}{\hbar}\frac{\hat{p}^2}{2m} t} \sin(k_n q_F) \cdot e^{-\frac{i}{\hbar} p q_i} \cdot \sum_{m=-\infty}^{\infty} \cdot e^{-2\pi i n m}$$

$$= \frac{2i}{L} \sum_{n=-\infty}^{\infty} e^{-\frac{iEt}{\hbar}} \cdot \sin(k_n q_F) \, e^{\frac{i}{\hbar} p q_i}$$

$$\doteq \mathrm{Re}\left(\frac{2}{L} \sum_{n=1}^{\infty} e^{-\frac{i}{\hbar} Et} \sin(k_n q_F) \sin(k_n q_i) \right)$$

Identities in the lines above:

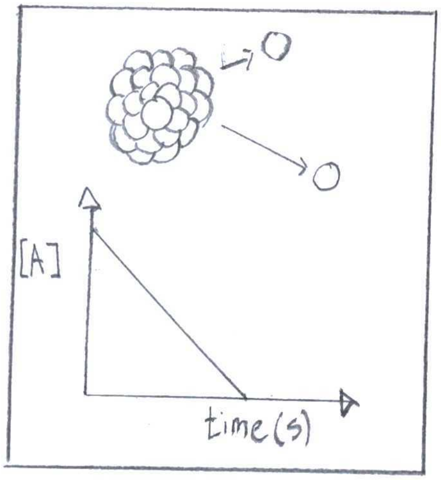
Momentum	Energy	Sine to exponential
$p = \frac{\pi n \hbar}{L}$	$E = \frac{(\hbar k_n)^2}{2m}$	$\operatorname{sinc}(\theta) = \frac{e^{i\theta} - e^{-i\theta}}{2i}$

(from book) "Poisson Summation Formula"

$$\sum_{m=-\infty}^{\infty} f(m) = \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} d\phi \, f(\phi) e^{2\pi i n \phi}$$

Exercise #7: (pg 118)

(from problem) "Effective Potential"



"Heavy nucleus having a finite rate of α-decay"

$$U(r) = 2(Z-1)e^2/r$$

Temperature:

$$\frac{3}{2} k_B T = U(r) - E$$

$$T = \frac{2}{3k_B}\left[U(r_0) - E\right]$$

(1850-1900)

Kinetic Energy in ideal gases

$$\boxed{K.E = \frac{3}{2} k_B T}$$

Mean Energy:

$$E_m = -\frac{d\ln Q}{d\beta}$$

$$= -\frac{d}{d\beta} \ln \sum_{n=1}^{\infty} e^{-E_a/\beta}$$

$$= -\frac{d}{d\beta} \ln\left(\frac{e^{-\beta E}}{1-e^{-\beta E}}\right)$$

$$= \frac{\beta e^{\beta E}}{e^{\beta E} - 1}$$

$$= \frac{\beta}{1-e^{-\beta E}}$$

$$= \frac{k_B T}{1-e^{-E/k_B T}}$$

$$= \frac{\frac{3}{2}(U(r_0)-E)}{1-e^{-3E/2(U(r_0)-E)}} = \frac{3(Z-1)e^2/r_0 - E}{2\left(1-e^{-3E/2(Z-1)e^2/r_0 - E}\right)}$$

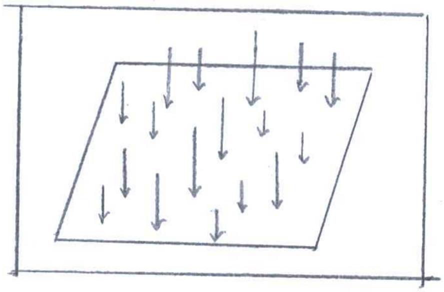
(1867)

Mean free energy

$$E = -\frac{d\ln Q}{d\beta}$$

where Q = canonical ensemble

Exercise #8
(pg 118)



"Uniform electric field ... applied to the surface of metal"

Cite: Fermi, Enrico

"Sulla quantizzazione del gas perfetto monoatomico"

Italy (1926)

Fermi limited the classical equation for energy ($\frac{3}{2} k_B T$) at zero degrees Kelvin. Other tid bits until 1962, then 1999 in laboratory.

Current from electron tunneling:

0K — Temperature — 293K

"Cold emission" — "Hot emission"

Tunneling Current ⟷ Fowler Nordeam Tunneling ⟷ Schotky emitter

$J = A e^{-B\Phi^{3/2}/E}$ $\quad$ $J \propto A E e^{-B\Phi^{3/2}/E}$ $\quad$ $J = A T^2 e^{-q(\Phi - \Delta\Phi)/k_B T}$

... when A, B = Constants specific to metal, shape, and temp.

Φ = Work function

E = Energy Fermi gas = $\frac{3}{2} k_B T$

Exercise in Edition no. 2 (pg 139)

(1.17) "Euler-Lagrange equation"

$$\forall x, i; \quad \frac{\partial \mathcal{L}}{\partial \phi^i(x} - \partial_\mu \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi^i(x)} = 0$$

$$\mathcal{L}(\theta, \phi) = \vec{B}\cos\theta + i(1-\cos\theta)\partial_\tau \phi$$

$$\frac{\partial \mathcal{L}}{\partial \theta} = i\mathring{\phi}\sin\theta$$

$$= B_x \cos\theta\cos\phi + B_y \cos\theta\sin\phi - B_z \sin\theta$$

$$\frac{\partial}{\partial \tau}\frac{\partial \mathcal{L}}{\partial \mathring{\phi}} = \mathring{\theta}$$

$$= B_x\cos\theta - B_y\cos\phi$$

<u>Block Function:</u>

IF $B = B_x + B_y + B_z$

and $n = \sin(\theta)\cos(\theta)\hat{x} + \sin\theta\sin\theta\hat{y} + \cos\theta\hat{z}$

$$i\partial_\tau n = i\mathring{n} -$$

$$= \begin{pmatrix} \mathring{\theta}\cos\theta\cos\phi - \mathring{\phi}\sin\theta\sin\phi \\ \mathring{\theta}\cos\theta\sin\phi + \mathring{\phi}\sin\theta\cos\phi \\ -\mathring{\theta}\sin\theta \end{pmatrix}$$

$$= \begin{pmatrix} i(B_x\sin\phi - B_y\cos\phi)\cos\theta\cos\phi - (B_x\cos\theta\cos\phi \\ \quad + B_y\cos\theta\sin\phi - B_z\sin\theta)\sin\phi \\ i(B_x\sin\phi - B_y\cos\phi)\cos\theta\sin\phi + i(B_x\cos\theta\cos\phi \\ \quad + B_y\cos\theta\sin\phi - B_z\sin\theta)\cos\phi \\ -i(B_x\sin\phi - B_y\cos\phi) \end{pmatrix}$$

$$= \begin{pmatrix} B_y \cos\theta - B_z \sin\theta \sin\phi \\ -B_x \cos\theta + B_z \sin\theta \cos\phi \\ B_x \sin\phi \sin\theta - B_y \sin\theta \cos\phi \end{pmatrix}$$

$$= -n \times \vec{B}$$

$$= B \times n$$

Exercise in edition no. 2. (pg 145)

(3.6) "Hamiltonian Path Integral"

$$\langle q_F | e^{-iHt/\hbar} | q_i \rangle = \int \mathcal{D}x \, e^{\left(\frac{i}{\hbar}\int_0^t dt' (p\dot{q} - H(p,q))\right)}$$

$$G(q_F, q_i, t) = \sum_{n=-\infty}^{\infty} \langle q_F | e^{-\frac{i}{\hbar}\frac{\hat{p}^2 t}{2m}} | q_i - 2na \rangle - \langle q_F | e^{-\frac{i}{\hbar}\frac{\hat{p}^2 t}{2m}} | q_i + 2na \rangle$$

$$= \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{e^{-\frac{i}{\hbar} p q_F}}{\sqrt{2\pi\hbar}} \cdot e^{-\frac{i}{\hbar}\frac{\hat{p}^2 t}{2m}} \frac{e^{\frac{i}{\hbar}(q_i + 2na)}}{\sqrt{2\pi\hbar}} \cdot dp - \int_{-\infty}^{\infty} \frac{e^{-\frac{i}{\hbar} p q_F}}{\sqrt{2\pi\hbar}} \cdot e^{-\frac{i}{\hbar}\frac{\hat{p}^2 t}{2m}} \frac{e^{\frac{i}{\hbar}(q_i + 2na)}}{\sqrt{2\pi\hbar}} \cdot dp$$

$$= \sum_{n=-\infty}^{\infty} \left(\frac{1}{2\pi\hbar}\right) \int_{-\infty}^{\infty} e^{-\frac{i}{\hbar}\left[\frac{\hat{p}^2 t}{2m} + \frac{(q_F - q_i + 2na)}{}\left(\frac{t}{t}\right)\right]} \cdot dp - \int_{-\infty}^{\infty} e^{-\frac{i}{\hbar}\left[\frac{\hat{p}^2 t}{2m} + \frac{(q_F - q_i - 2na)}{}\left(\frac{t}{t}\right)\right]} \cdot dp$$

$$= \frac{m}{2\pi i \hbar t} \sum_{n=-\infty}^{\infty} e^{\frac{im(q_F - q_i + 2na)^2}{2\hbar t}} - e^{\frac{im(q_F + q_i - 2na)^2}{2\hbar t}}$$

Exercise #9:
(pg 146)

(3.71) "Field Integral"

$$Z = \int D\psi \, e^{-S[\phi]} \quad , \quad S[\phi] = \int_0^\beta d\tau \left(\bar{\psi} \partial_\tau \psi + H(\bar{\psi}, \psi) \right)$$

$$\text{where } D\psi = \lim_{N \to \infty} \prod_{n=1}^{N} d\psi^n$$

(3.69) "Hamiltonian for a two body interaction"

$$\hat{H}(a^\dagger, a) = \sum_{ij} h_{ij} a_i^\dagger a_j + \sum_{ijkl} V_{ijkl} a_i^\dagger a_j^\dagger a_k a_l$$

(from Problem) "Transformation"

$$\psi = e^{i\phi} \psi \quad , \quad \bar{\psi} = \bar{\psi} e^{-i\phi}$$

$$S[e^{i\phi}\psi] = \int_0^\beta d\tau \left(\bar{\psi} e^{-i\phi} \partial_\tau \psi e^{i\phi} + H(\bar{\psi} e^{-i\phi}, \psi e^{i\phi}) \right)$$

$$= \int_0^\beta d\tau \left(\bar{\psi} e^{-i\phi} \partial_\tau \psi e^{i\phi} + \sum_{ij} h_{ij} \bar{\psi} e^{-i\phi} \psi e^{i\phi} \right.$$

$$\left. + \sum_{ijkl} V_{ijkl} \bar{\psi}_i e^{-i\phi} \bar{\psi}_j e^{-i\phi} \psi_k e^{i\phi} \psi_l e^{i\phi} \right)$$

$$= \int_0^\beta d\tau \left(\bar{\psi} \partial_\tau \psi + H(\psi, \psi) \right)$$

$$= S[\Psi]$$

<u>Noether current:</u>

(1.43) "Noethers Theorem

$$j_\mu = \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^i)} \partial_\nu \phi^i - \mathcal{L}\delta_{\mu\nu} \right) \frac{\partial x_\nu}{\partial w_a} \bigg|_{w=0} - \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^i)} F_a^i$$

where $\mathcal{L} = \bar{\Psi}\partial_\tau \Psi + H(\bar{\Psi}, \Psi)$

$$j_\mu = \bar{\Psi} e^{-i\phi} \circ i \Psi e^{i\phi} - i \bar{\Psi} e^{-i\phi} \circ \Psi$$

$$= 0$$

Exercise #10: (pg 147)

(3.89) "Invariant Transformation"

$$\Psi \to e^{i\phi_a \gamma^5} \circ \Psi \quad , \quad \bar{\Psi} \to \bar{\Psi} e^{-i\phi_a \gamma^5}$$

where $[\gamma^\mu, \gamma^5] = 0$ for $\mu = 0, 1$

The invariant transformation describes a real world observable with equal values in 2π-increments, such as interference by a wave. A function for the wave is

$$\Psi(x) = e^{i(kx - wt)} \circ$$

(3.86) "One-dimensional electron gas"

$$S_0[\psi] = \int d^2x\, \psi^\dagger(\sigma_0 \partial_{x_0} - i\sigma_3 \partial_{x_1})\psi$$

$$= \int d^2x\, \psi(\sigma_1 \partial_{x_1} - i\sigma_2 \partial_{x_1})\psi$$

(from problem) "Transformation"

$$\psi \to e^{i\phi_a\sigma_3}\psi \quad , \quad \bar{\psi} \to \psi^\dagger e^{-i\phi_a\sigma_3}$$

$$S_0[e^{i\phi\sigma_3}\psi] = \int d^2x\, \psi^\dagger e^{-i\phi_a\sigma_3}(\sigma_1\partial x_0 - \sigma_2\,\partial x_1)\psi e^{i\phi_a\sigma_3}$$

$$= \int d^2x\, \psi^\dagger(\sigma_1\partial x_0 - \sigma_2\partial x_1)\psi$$

$$= S[\psi]$$

$$S_0[\psi^\dagger e^{-i\phi\sigma_3}] = \int d^2x\, \psi^\dagger e^{-i\phi_a\sigma_3}(\sigma_1\partial x_0 - \sigma_2\partial x_1)\psi$$

$$= S[\psi] \quad \ldots \text{with } \phi_a = 2\pi$$

$$S_0[\psi e^{i\phi_a\sigma_3}] = \int d^2x\, \psi^\dagger(\sigma_1\partial x_0 - \sigma_2\partial x_1)\psi e^{i\phi_a\sigma_3}$$

$$= S[\psi] \quad \ldots \text{with } \phi_a = 2\pi$$

Exercise #11: (pg 149)

(3.93) "Anticommutator identity"

$$[e^{is\hat{\theta}(x)} e^{i\phi(x)}, e^{is'\theta(x')} e^{i\hat{\phi}(x')}]$$

$$\propto e^{i\pi\Theta(x'-x)} + e^{i\pi\Theta(x-x')} = 0$$

(3.92) "String operator"

$$[\hat{\phi}(x), \Theta(y)] = -i\pi\Theta(y-x)$$

Commutator identities

$$e^{\hat{A}} \circ \hat{B} \circ e^{-\hat{A}} = e^{[\hat{A},]} \circ \hat{B}$$

$$e^{\hat{A}} \cdot e^{\hat{B}} \cdot e^{\hat{A}} = e^{e^{\hat{A}} \circ \hat{B} e^{-\hat{A}}}$$

$$[e^{iA} \cdot e^{iB}, e^{iA'} e^{iB'}] =$$

$$= e^{iA} \cdot e^{iB} \circ e^{iA'} \circ e^{iB'} + e^{iA'} \cdot e^{iB'} \circ e^{iA} \circ e^{iB}$$

$$= e^{-[\hat{\Phi}(x), \hat{\Theta}(x)]} + e^{[\hat{\Phi}(x), \hat{\Theta}(x')]}$$

$$= e^{i\pi\Theta(x'-x)} + e^{i\pi\Theta(x-x')}$$

where $[\hat{\Phi}(x), s\hat{\Theta}(x)] = -i\pi\Theta(x-x')$

Canonical Anticommutation relation:

$$[c_i, c_i^{\dagger}] = \frac{1}{\sqrt{2}} e^{is\theta_i} \cdot e^{i\phi_i} \circ \frac{1}{\sqrt{2}} e^{-i\phi_i} \circ e^{-is\theta_i}$$

$$+ \frac{1}{\sqrt{2}} e^{-i\phi_i} e^{-is\theta_i} \cdot \frac{1}{\sqrt{2}} e^{is\theta_i} \circ e^{i\phi_i}$$

$$= \frac{1}{2}\left(e^{-[s\theta_i, \phi_i]} + e^{-[s\theta_i, \phi_i]}\right)$$

$$= e^{i\pi\theta}$$

$= 1$ $\therefore$ when $[s\theta_i, \phi_i] = \delta_{ij}$

$= 1$

Exercise #12: (pg 151) "Lagrangian from acting integral"
(pg 151)

$$S_M[\theta] = \frac{1}{2\pi}\int d\tau\, dx\left((\partial_\tau\theta)^2 - (\partial_x\theta)^2\right)$$

The book wants a "quick" alternative.

$(\partial_x\phi + i\partial_\tau\theta)^2 = 0$, so $(\partial_\tau\theta)^2 = -(\partial_x\phi)^2 - 2i\partial_\tau\partial_x\phi$

$$S_M[\theta] = \frac{1}{2\pi}\int d\tau\, dx\left(-(\partial_x\phi)^2 - 2i\partial_\tau\partial_x\phi - (\partial_x\theta)^2\right)$$

$$= -S[\theta]$$

$$S[\theta] = \frac{1}{2\pi}\int d\tau\, dx\left((\partial_x\phi)^2 + 2i\partial_\tau\partial_x\phi + (\partial_x\theta)^2\right)$$

Cite: СТРАТОНОВИЧ, Р. Л.
"ОБ ОДОМ МЕТОДЕ ВЫЧИСЛЕНИЯ КВАНТОВІХ ФУНКЦИИ РАСПРЕДЕЛЕНИЯ"
ДОКЛАДЫ АКАДЕМИИ НАУК, СССР
(1957)

J. Hubbard (1959) describes СТРАТОНОВИЧ as a solution to Bose-Einstein condensates by chemical potential. The syntax matches, but without bounds.

Exercise #13: (pg 152) (1.43) "Noethen Current"

$$j^{\kappa\mu} = \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^i)}\partial_\nu \phi^i - \mathcal{L}\delta^\mu_\nu\right)\frac{\partial x^\nu}{\partial w_a}\Big|_{w=0} - \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^i)} F_a^i(\phi)$$

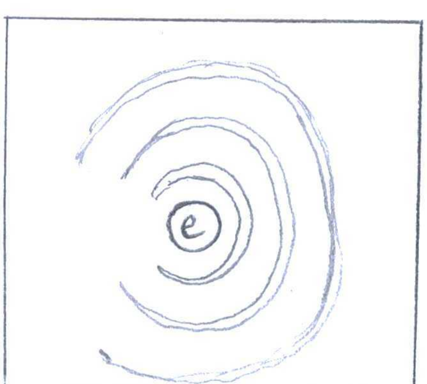
$$= \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^i)} 2\mu$$

$$j_v^0 = \rho \qquad j_v^1 = -i \qquad j = \rho_+ - \rho_- \text{ (pg 151)}$$

$$= \frac{\partial_v \theta}{\pi} \qquad = \frac{-i\partial_x \phi}{\pi} \qquad = \frac{1}{2\pi}(\partial_x\theta + \partial_x\phi) - \frac{1}{2\pi}(\partial_x\theta - \partial_x\phi)$$

$$= \frac{\partial_x \Phi}{\pi}$$

Exercise #14: (pg 163) (from problem) "Density-Density Correlation function"



"Friedel oscillation

$$\Pi(x,t) = \langle \hat{\rho}(x,t)\rho(0,0)\rangle - \langle\hat{\rho}(x,t)\rangle\langle\hat{\rho}(0,0)\rangle$$

where $\hat{\rho}$ = "Groundstate" $= a^\dagger a$

$$a(x) = e^{ik_F x} a_+(x) + e^{-ik_F x} a_-(x)$$

"Left-and-right wave"

(from problem) "Von neumann equation"

$$\mathring{a}_{sq} = i[\hat{H}, a_{sq}]$$

$$\hat{H} = \sum_{q,s} V_F(p_F + sq) a_{sq}^\dagger a_{sq}$$

Time dependent operator:

$$\mathring{a}_{sq} = \frac{da}{dt}$$

$$= i[\hat{H}, a_{sq}]$$

$$= i[\hat{H} a_{sq} - a_{sq} \hat{H}]$$

$$\frac{da}{a_{sq}} = -i V_F(p_F + s_q)\, dt$$

$$a_{sq} = e^{-iV_F(p_F + s_q)t}$$

The static response decays with a maximum $|X|^{-1}$ where the expectation (average) is $1/x^2$ from the correlation function.

$$\Pi(x,t) = \frac{1}{4\pi}\left(\frac{1}{(x-v_F t)^2} + \frac{1}{(x+v_F t)^2} + \frac{2\cos(2p_F x)}{x^2-(v_F t)^2}\right)$$

Problem 3.8.1 - Quantum Harmonic Oscillator:

a) (3.71) "Field integral"

$$Z = \int D\Psi e^{-S[\Phi]} \quad , \quad S[\phi] = \int_0^\beta d\tau \left(\bar{\Psi}\partial_\tau \Psi + H(\bar{\Psi}, \Psi)\right)$$

(pg 102) "Quantum Mechanical amplitude"

$$Z = \int dq \langle q | e^{-it\hat{H}/\hbar} | q \rangle \Big|_{\substack{\hbar = 1/\beta \\ t = -iL}}$$

(from problem) "Hamiltonian"

$$\hat{H} = \hat{p}^2/2m + m\omega^2\hat{q}^2/2$$

$$\text{So,} \quad S[q] = \int_0^t dt\, \hat{H}$$

$$= \int_0^t dt \left(\frac{\hat{p}^2}{2m} + \frac{m\omega^2\hat{q}^2}{2}\right)$$

$$= \int_0^t dt' \frac{m}{2}\left(\dot{q}^2 - \omega^2 q^2\right)$$

$$= \int_0^t dt \frac{m}{2}\left(\left(\frac{d}{dt}(A\sin(\omega t) + B\cos(\omega t)\right)^2 - \omega^2\left(A\sin(\omega t) + B\cos(\omega t)\right)^2\right)$$

$$= \frac{m\omega}{4}\left[(A^2-B^2)\sin(2\omega t) - 4AB\sin^2(\omega t)\right]$$

$$= \frac{m\omega^2}{2}\left[(A^2-B^2)\frac{\sin(2\omega t)}{2\omega} + \frac{2AB\cos(2\omega t)-1}{2\omega}\right]$$

$$\boxed{\begin{array}{l}\underline{\text{Identity:}} \\ \sin^2(\omega t) = \frac{1-\cos(2\omega t)}{2}\end{array}}$$

$$= \frac{m\omega}{2}\left[(q_i^2+q_F^2)\cot(\omega t) - \frac{2q_i q_F}{\sin(\omega t)}\right]$$

for $A = q_i$ and $B = q_F$

$$\langle q_F | e^{-iHt/\hbar} | q_i \rangle = \int_{q[0]=q_i}^{q[t]=q_F} Dq\, e^{iS[q]/\hbar}$$

$$= \int_{q[0]=q_i}^{q[t]=q_F} Dq \cdot e^{\frac{i}{\hbar}\frac{m\omega}{2}\left[(q_i^2+q_F^2)\cot(\omega t) - 2q_i q_F/\sin(\omega t)\right]}$$

$$= \left(\frac{m\omega}{2\pi i\hbar\sin(\omega t)}\right) e^{\frac{i}{2\hbar}m\omega\left((q_F^2+q_i^2)\cot(\omega t) - 2q_i q_F/\sin(\omega t)\right)}$$

The singularities at $t = n\pi/\omega$ for $n = 1, 2, \ldots$ associate to uniform spring models. The oscillator is in ground state.

b) Wavepacket : $\Psi(q, t=0) = (2\pi a)^{-1/4} \circ \exp[-q^2/4a]$

$$\Psi(q) = J(t) \int_{-\infty}^{\infty} dq' \frac{1}{(2\pi a)^{1/4}} e^{\frac{i}{\hbar}\frac{m\omega}{2}\left[(q_F^2+q_i^2)\cot(\omega t) - \frac{2q_i q_F}{\sin(\omega t)}\right]}$$

$$= J(t) \int_{-\infty}^{\infty} dq' \frac{1}{(2\pi a)^{1/4}} e^{\beta^2/2\alpha} \circ e^{\left(\frac{i}{2\hbar} m\omega q^2 \cot(\omega t)\right)}$$

where $\beta = im\omega q / \hbar \sin(\omega t)$

$\alpha = 1/2a - im\omega\cot(\omega t)/\hbar$

$$= J(t) \frac{1}{(2\pi a)^{1/4}} \sqrt{\frac{2\pi}{\alpha}}\, e^{\beta^2/2\alpha} \circ e^{\left(\frac{i}{2\hbar} m\omega q^2 \cot(\omega t)\right)}$$

where $\beta^2/2\alpha = -(1 + ik\cot(\omega t) q^2/4a(t))$

$$= \frac{1}{(2\pi a(t))^{1/4}} e^{-q^2/4a(t)} \circ e^{i\phi(q,t)}$$

<u>Width:</u>

$a(t) = a(\cos^2(\omega t) + K^{-2}\sin^2(\omega t))$

where $K = 2am\omega/\hbar$

c) Initial wave packet:

$$\Psi(a, t=0) = (2\pi a)^{-1/4} \circ e^{\frac{i}{\hbar} mvq - \frac{1}{4a} q^2}$$

<u>Wavepacket at t:</u>

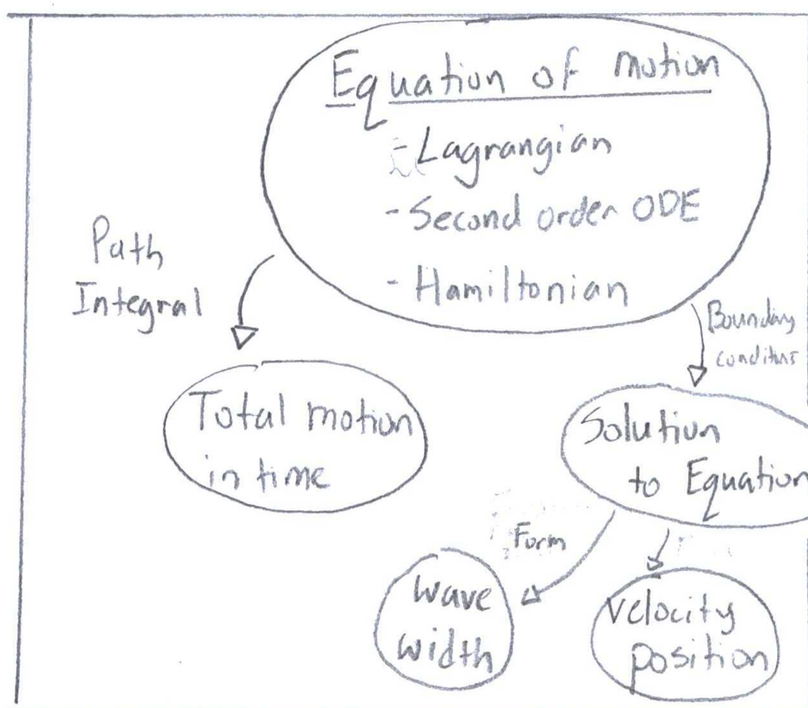
000 b

000 same as b, except $\beta = \frac{i}{\hbar} \frac{m\omega}{\sin(\omega t)} \times \left(q - \frac{v}{\omega}\sin(\omega t)\right)$

Position: $q_0(t) = (v/\omega)\sin(\omega t)$

velocity: $v(t) = v_0 \cos(\omega t)$

Problem 3.8.2: Density Matrix:



$$\rho(q,q') = \langle q_f | e^{-i\hat{H}t/\hbar} | q_i \rangle$$

$$= \langle q | e^{-\beta H} | q' \rangle$$

$$= \langle q | e^{-(i/\hbar)\beta H(\hbar/i)} | q' \rangle$$

$$= \left(\frac{m\omega}{2\pi i \hbar \sin(i\beta\hbar\omega)}\right) e^{\frac{i}{2\hbar} m\omega\left((q_i^2+q_f^2)\cot(\beta\hbar\omega) - \frac{2q_i q_f}{\sin(i\beta\hbar\omega)}\right)}$$

Complex identities

$i\sin(x) = \sin(ix)$

$i\cot(x) = \coth(x)$

$\sinh(x) = i\sin(ix)$

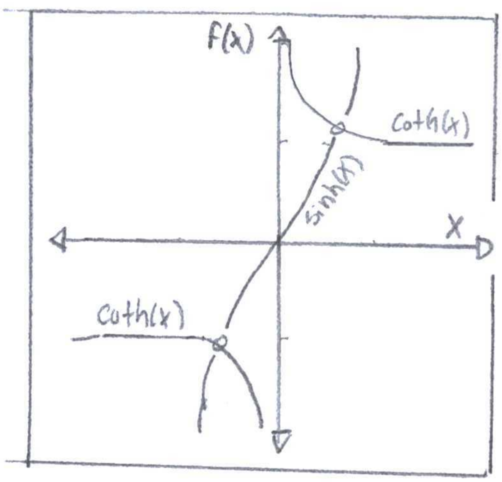
$$= \left(\frac{m\omega}{2\pi\hbar\sinh(\beta\hbar\omega)}\right) e^{-\frac{m\omega}{2\hbar}\left((q^2+q'^2)\coth(\beta\hbar\omega) - \frac{2qq'}{\sinh(\beta\hbar\omega)}\right)}$$

i) $T \ll \hbar\omega$ (Low Temp):

$$\rho(q,q') = \left(\frac{m\omega}{\pi\hbar\, e^{\beta\hbar\omega}}\right)^{1/2} e^{-\frac{m\omega}{2\hbar}(q^2+q'^2)}$$

$=$ "Ground state"

$$= \langle q | n=0 | e^{-\beta E_0} | n=0 | q' \rangle$$



ii) $T \gg \hbar\omega$ (High Temp):

$$\rho(q,q') = \left(\frac{m}{2\pi\beta\hbar^2}\right)^{1/2} e^{-m(q-q')^2/2\beta\hbar^2} \cdot e^{-\beta m\omega^2(q^2+q'^2+qq')/6}$$

$$\cong e^{-\beta m\omega^2 q^2/2} \quad \text{when} \quad q-q'=0$$

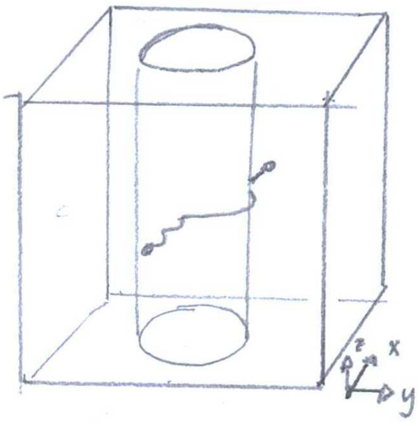
"Maxwell Boltzmann distribution"

Taylor series

$\coth(x) = \frac{1}{x} + \frac{x}{3} + \dots$

$\sinh^{-1}(x) = \frac{1}{x} - \frac{6}{x}$

Problem 3.8.3: Depinning transition and bubble nucleation:



"A flux line in a superconductor with a columnar defect"

(from problem) "Euclidean-time action"

$$S[u] = \int dz\,d\tau \left(\frac{1}{2}\rho \mathring{u}^2 + \frac{1}{2}\sigma(\partial_z u)^2 + V(|u|) \right)$$

Where $u(z,\tau)$ = String dispacement in xy-plane

ρ = density per unit length

σ = tension in the string

(from problem) "External in-plane field"

$$S_{ext} = -f \int dz\,d\tau \; u \circ e_x$$

a) Saddle-point in $z\tau$-space:

Rescaling: "Dimensionalization by Strogatz"

$$F_{superconductor} = F_{external}$$

$$\frac{1}{2}\rho \mathring{u}^2 + \frac{1}{2}\sigma(\partial_z u)^2 + V(|u|) = -f u \circ e_x$$

$$\rho \ddot{u} + \sigma \partial_z^2 u_x + V'(u)\frac{u_x}{u} = -f$$

If $\tau = (\rho/\sigma)^{1/4}\tilde{\tau}$ and $z = (\sigma/\rho)^{1/4}\tilde{z}$

$$\sqrt{\sigma}\,\rho\,\partial^2 u_x = -f - V'(|u|)\,\frac{u_x}{u}$$

... when $\partial^2 = \partial_\tau^2 + \partial_z^2$

<u>Roots:</u>

$$u_x(r) = \begin{cases} 0 & r > R \quad \text{"Zero potential outside column"} \\ g(r) & r < R \quad \text{"finite potential near column"} \end{cases}$$

b) Thin-wall approximation: $V'(|u|) = 0$

① <u>Boundary conditions:</u>

$u_x(r > R) = 0$

$u_x(r < R) = g(r)$

② <u>Equation:</u>

$$\sqrt{\rho\sigma}\,\ddot{u}_x + f = 0$$

③ <u>Solution:</u>

$$\sqrt{\rho\sigma}\,\ddot{g}(r) = -f$$

$$\ddot{g}(r) = -f/\sqrt{\rho\sigma}$$

$$\dot{g}(r) = -f/\sqrt{\rho\sigma}\cdot r$$

$$g(r) = \frac{(R^2 - r^2)\,f}{\sqrt{4\sigma\rho}}$$

c) Probability of detaching:

$$\Delta S_{saddle} = \int_0^R d^2 r \left(\frac{\sqrt{\sigma\rho}}{2} (\partial g(r))^2 + V_0 - f \circ u \right)$$

$$= -\pi\, 4 \cdot R^3 \left(\frac{3f^2}{16(\rho\sigma)^{1/2}} - V_0 \right)$$

$$= 0$$

$$R^3 = \frac{4 V_0 \sqrt{\sigma\rho}}{3f^2}$$

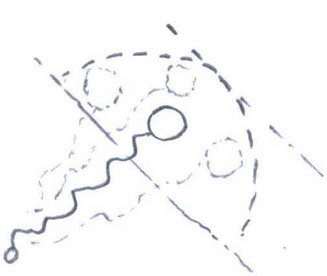
Typeset in book is incorrect by R^2 versus R^3.

Tunneling rate:

$$W \propto e^{-S[R]}$$

$$= e^{-4\pi V_0 \sqrt{\rho\sigma} / 3f^2}$$

Problem 3.8.4 - Tunneling in a dissipative environment:



"Particle.. coupled to a quantum mechanical string"

a) (From problem) "Sinusoidal potential"

$$U(q) = 2g \sin^2(\pi q / q_0)$$

(from problem) "Euclidean-time Path Integral"

$$\mathcal{Z} = \int Dq\, e^{-S_{part}[q]}, \quad S_{part}[q] = \int_{-\infty}^{\infty} d\tau \left(\frac{m}{2} \dot{q}^2 + U(q) \right)$$

(from problem) "Boundary conditions"

$$q(\tau=-\infty)=0 \quad , \quad q(\tau=\infty)=q_0$$

(from problem) "Instanton trajectory"

$$q_{el}(\tau)=(2q_0/\pi)\arctan(e^{w_0\tau})$$

where $w_0=(2\pi/q_0)\sqrt{g/m}$

$$S[q_0]=\int_0^\infty d\tau\left(\frac{m}{2}\dot{q}_{el}^2+U(q_{el})\right)$$

$$=\int_0^\infty d\tau\, m\dot{q}_{el}^2$$

$$=m\int_0^\infty dq\,\dot{q}_{el}^2$$

$$=2\frac{mq_0^2}{\pi^2}w_0$$ ... where $\arctan(e^{w_0\tau}) \cong w_0$

b) (from problem) "Path Integral"

$$Z=\int Du\int Dq\,\delta(q(\tau)-u(\tau,x=0))e^{-S_{String}[u]-S_{Part}[q]}$$

(from problem) "Classical action of the String"

$$S_{String}[u]=\frac{1}{2}\int_{-\infty}^{\infty}d\tau\int_{-\infty}^{\infty}dx(\rho\dot{u}^2+\sigma(\partial_x u)^2)$$

(from problem) "Functional δ-function"

$$\delta(q(\tau)-u(\tau,0)) = \int Df \exp\left(i\int_{-\infty}^{\infty} d\tau\, f(\tau)(q(\tau)-u(\tau,0)\right)$$

$$S_{string} = \frac{1}{2}\int_{-\infty}^{\infty} d\tau \int_{-\infty}^{\infty} dx \left(\rho \ddot{u} + \sigma(\partial_x u)^2\right)$$

$$= \frac{1}{2}\int_{-\infty}^{\infty}\frac{dw}{2\pi}\int_{-\infty}^{\infty}\frac{dk}{2\pi}(\rho w^2 + \sigma k^2)\,|u(w,k)|^2$$

$$\delta(q(\tau)-u(\tau,0)) = \int Df \exp\left(i\int_{-\infty} d\tau\, f(\tau)(q(\tau)-u(\tau,0)\right)$$

$$= \int Df \exp\left(i\int_{-\infty}^{\infty}\frac{dw}{2\pi} f(w)\left(q(-w) - \int\frac{dk}{2\pi}u(-w,-k)\right)\right)$$

$$= \int Du\, e^{-S_{string} \int d\tau\, f(\tau)\, u(\tau,0)}$$

$$\propto e^{-\frac{1}{2}\int_{-\infty}^{\infty}\frac{dw}{2\pi}\int_{-\infty}^{\infty}\frac{dk}{2\pi}\frac{1}{\rho w^2 + \sigma k^2}|f(w)|^2}$$

$$\propto e^{-\frac{1}{2}\int_{-\infty}^{\infty}\frac{dw}{2\pi}\circ\sqrt{\rho w^2 \sigma}\,|f(w)|^2}$$

$$\propto e^{-\frac{\eta}{2}\int_{-\infty}^{\infty}\frac{dw}{2\pi}|w|\,|f(w)|^2}$$

... when $\eta = \sqrt{4\rho\sigma}$

$$\delta(q(\tau)-u(\tau,0))=\int Du\, e^{-i\int d\tau f(\tau)u(\tau,0)-\frac{\eta}{2}\int_{-\infty}^{\infty}\frac{dw}{2\pi}|w||f(w)|^2}$$

$$=\int Du\, e^{-S_{eff}[q]}$$

where $S_{eff}[q]=S_{part}+\frac{\eta}{2}\int\frac{dw}{2\pi}|w||f(w)|^2$

c) (from problem) "Instanton - Anti-instanton pair"

$$q(\tau)=q_{cl}(\tau+\bar{\tau}/2)-q_{cl}(\tau-\bar{\tau}/2)$$

$$q(w)=\int_{-\bar{\tau}/2}^{\bar{\tau}/2}d\tau\, e^{iw\tau}$$

$$=q_0\bar{\tau}\sin(w\bar{\tau}/2)/(w\bar{\tau}/2)$$

$$S_{eff}=S_{part}[q]+\frac{\eta}{2}\int\frac{dw}{2\pi}|w||f(w)|^2$$

$$=S_{part}[q]+\frac{\eta}{2}\int\frac{dw}{2\pi}|w|(q_0\bar{\tau})^2\frac{\sin^2(w\bar{\tau}/2)}{(w\bar{\tau}/2)^2}$$

$$=S_{part}[q]+\frac{\eta q_0^2}{\pi}\ln(w_0\bar{\tau})$$

d) <u>Typical separation of an instanton anti-instanton pair:</u>

$$\langle\bar{\tau}\rangle=\int d\tau\cdot\tau\, e^{-S_{eff}[q]}$$

$$= \int d\tau \cdot \tau e^{-S_{part}[q]} \cdot e^{-\frac{\eta}{\pi} q_0^2 \ln(\omega_0 \tau)}$$

$$= e^{-S_{part}[q]} \cdot \int d\tau \cdot \tau \cdot e^{-\frac{\eta}{\pi} q_0^2 \ln(\omega_0 \tau)}$$

$$= e^{-S_{part}[q]} \cdot \int d\tau \cdot \tau \cdot e^{\ln\left[(\omega_0 \tau)^{-\frac{q_0^2 \eta}{\pi}}\right]}$$

$$= e^{-S_{part}[q]} \cdot \int d\tau \cdot \tau \, (\omega_0 \tau)^{-\frac{q_0^2 \eta}{\pi}}$$

$$= e^{-S_{part}[q]} \cdot \omega_0^{-\frac{q_0^2 \eta}{\pi}} \cdot \int d\tau \cdot \tau^{1-\frac{q_0^2}{\pi}\eta}$$

If the exponent defines duration to traversal by a particle. A positive exponent describes long-time, while a negative exponent describes small time separation.

Problem 3.8.5 - Exercises on Fermionic states:

a) $\langle \eta | a_i^\dagger = \langle 0 | e^{-\sum a_j n_j} \cdot a_i^\dagger$

$$= \langle 0 | \prod_j (1 - a_j n_j) a_i^\dagger$$

$$= \langle 0 | \underbrace{(1 - a_i \eta_i)}_{\text{"Sketchy"}} a_i^\dagger \prod_{j \neq i} (1 - a_j \eta_j)$$

$$= \langle 0 | a_i a_i^\dagger \eta \prod_{j \neq i} (1 - a_j \eta_j)$$

$$= \langle 0 | \prod (1 - a_j \eta_j) \eta_j$$

$$= \langle \eta | \bar{\eta}_i$$

b) $a^\dagger | \eta \rangle = -\partial_\eta a_i^\dagger | \eta \rangle$

c) $\langle \eta | v \rangle = \langle \eta | \prod_i (1 - v_i a_i^\dagger) | 0 \rangle$

$$= \langle \eta | \prod (1 + \eta^\dagger v) | 0 \rangle$$

$$= e^{\sum \eta_i v_i}$$

d) $\int d\eta \, e^{-\eta \bar{\eta}} | \eta \rangle \langle \eta$

(pg 131) "Schur's Lemma"

"Operators commute with all $\{a_i, a_i^\dagger\}$, and must be proportional to a unit...."

$$\int d\eta\, e^{-\eta\bar{\eta}}|\eta\rangle\langle\eta$$

$$\langle 0|\int d\eta\, e^{-\eta\bar{\eta}}|\eta\rangle\langle\eta|0\rangle = \int d\eta\, e^{-\bar{\eta}\eta}$$

$$= \langle 0|0\rangle$$

$$= 1$$

e) If $|n\rangle = a_1^\dagger \cdots a_n^\dagger|0\rangle$

and $\langle n| = \langle 0|a_n \cdots a_1$

then, $\langle\psi|n\rangle = \bar{\psi}_1, \cdots \bar{\psi}_n$

$$\langle n|\psi\rangle\langle\psi|n\rangle = (\psi_n,\cdots,\psi_1)(\bar{\psi}_1,\cdots,\bar{\psi}_n)$$

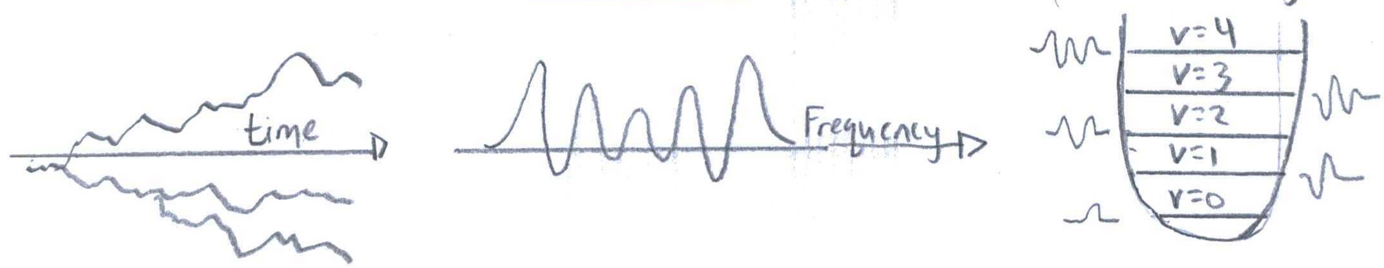
$$= \psi_1\bar{\psi}_1, \cdots, \psi_n, \bar{\psi}_n$$

$$= (\zeta\bar{\psi}_1\psi_1)\cdots(\zeta\bar{\psi}_n\psi_n)$$

$$= (\zeta\bar{\psi}_1)\psi_1\cdots(\zeta\psi_n)\psi_n$$

$$= \langle\zeta\psi|n\rangle\langle n|\psi\rangle$$

Problem 3.8.6 – Feynmann path integral from a functional field integral:



$$\begin{matrix}\text{Possible} \\ \text{States}\end{matrix} = Z = \int D\phi\, e^{-\int dt \hat{H}} \longleftrightarrow \text{Energy} \propto = \propto \hbar\omega(a^\dagger a + 1/2) \longleftrightarrow \text{Harmonic Oscillator}$$

Field integral with Harmonic Oscillator:

(from problem) "Harmonic oscillator Hamiltonian"

$$\hat{H} = \omega(a^\dagger a + 1/2)$$

(3.71) "Field Integral"

$$Z = \int D\psi\, e^{-S[\psi]}, \quad S[\psi] = \int_0^\beta d\tau\,(\bar{\psi}\partial_\tau \psi + H(\bar{\psi}, \psi))$$

$$= e^{-\beta\omega/2} \cdot \int D\phi\, e^{-\int_0^\beta d\tau\,(\bar{\phi}\partial_\tau \phi + \omega\bar{\phi}\phi)}$$

When $\phi(\tau) = \left(\frac{m\omega}{2}\right)^{1/2}\left(q(\tau) + \frac{i}{m\omega}p(\tau)\right)$

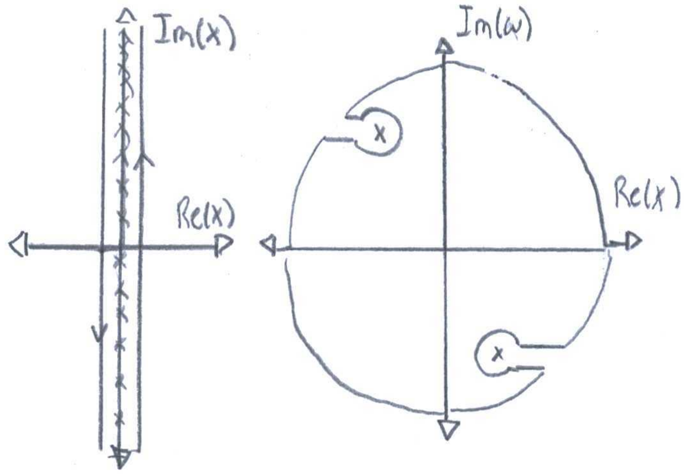
$$\bar{\phi}(\tau) = \left(\frac{m\omega}{2}\right)^{1/2}\left(q(\tau) - \frac{i}{m\omega}p(\tau)\right)$$

$$Z = e^{-\frac{\beta\omega}{2}}\int D\phi\, e^{-\int_0^\beta d\tau\left(\frac{p^2}{2m} + \frac{m\omega^2}{2}q^2 - (m\omega q - ip)\frac{\dot{q}}{2}\right)}$$

$\therefore$ when $\dot{p} = 0$

$\therefore$ when $\dot{p}$ and $\dot{q}$ are from Hamiltons equation in Classical mechanics.

Problem 3.8.7 - Quantum Partition Function in a Harmonic Oscillator:



"Matsubara Frequency"
(1955)

Partition function for Bosons:

(3.101) "Quantum partition function for an oscillator"

$$Z = \int D\phi \exp\left(-\int_0^\beta d\tau \left(\bar{\Phi}\partial_\tau\phi + \omega\bar{\Phi}\phi\right)\right)$$

(3.78) "Partition function decoupled"

$$Z = \prod_n \det\left(\beta(-i\omega + \hbar\omega)\right)^{-\xi}$$

where $\xi = \begin{cases} 1 & \text{bosons} \\ -1 & \text{fermions} \end{cases}$

$$= \prod_{n=1}^{\infty} \frac{1}{\left(\frac{2\pi n}{\beta}\right)^2 - \omega^2}$$

$$= \prod_{n=1}^{\infty} \frac{1}{\left(\frac{2\pi n}{\beta}\right)^2} \frac{1}{1 - \left(\frac{\beta\omega}{2\pi n}\right)^2}$$

$$= \prod_{n=1}^{\infty} \frac{1}{1 - \left(\frac{\beta\omega}{2\pi n}\right)^2}$$

Akin to an ansatz, the function equals zero.

$$= \sinh(\beta\omega/2)$$

... at low temperatures,

$$\lim_{T\to 0} Z_\beta = \lim_{T\to 0} \sinh(\beta\omega/2)$$

$$= 0$$

Euler-Wallis Product:

$$\frac{\sin(x)}{x} = \prod_{n=1}^{\infty}\left(1 - \frac{x^2}{n^2\pi^2}\right)$$

$$= \sinh(x)$$

Method of Mathematical physics:

$$\frac{\sin(\pi z)}{\pi z} = \prod_{n=1}^{\infty}\left(1 - \frac{z^2}{n^2}\right)$$

Partition function for fermions:

$$Z = \prod_{n=-\infty}^{\infty} \det(\beta(-i\omega + \hat{H}))$$

$$= \prod_{n=-\infty}^{\infty} \det(\beta(-i\omega + \hbar\omega))$$

$$= \prod_{n=-\infty}^{\infty}\left(1 + \left(\frac{\beta\omega}{(2n+1)\pi}\right)^2\right)$$

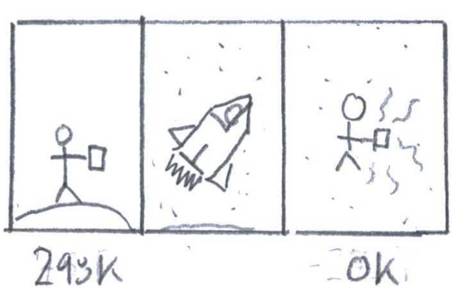
$$= \prod_{n=0}^{\infty}\left(1 + \left(\frac{\beta\omega}{(2n-1)\pi}\right)^2\right)$$

cite: Roy. R, Oliver F.

Digital library of mathematical functions

$$\cosh(z) = \prod_{n=1}^{\infty}\left(1 + \frac{4z^2}{(2n-1)^2\pi^2}\right)$$

Problem 3.8.8 - Frequency Summations:



(from problem) "Cooper pair propagator"

$$X^c_{\omega,q} = -\frac{T}{L^d}\sum_{\omega_m, p} G_0(i\omega_m, p)\,G_0(i\omega_n - i\omega_m, q-p)$$

where $G_0(p, i\omega_m) = (i\omega_m - \xi_p)^{-1}$

(pg 143) "Contour"

$$\sum h(\omega_n) = \xi \, \mathrm{Re}(g(z) \circ h(-iz))$$

$$= \frac{1}{e^{\beta\xi} - \zeta} \quad \text{"Fermions"}$$

(3.80) "Fermions"

$$g(z) = \frac{\beta}{e^{\beta z} + 1}$$

$$\chi^c_{\omega, q} = -\frac{T}{L^d} \sum_{\omega_m, p} G_0(i\omega_m, p) \circ G_0(i\omega_n - i\omega_m, q - p)$$

$$= -\frac{1}{L^d} \sum_p \sum_{\omega_m} h(\omega)$$

$$= -\frac{1}{L^d} \sum_p \frac{-n_F(\xi)}{i\omega - \xi_{p_1} - \xi_{p_2}} + \frac{-n_F(\xi - i\omega_m)}{i\omega - \xi_{p_1} - \xi_{p_2}}$$

$$= -\frac{1}{L^d} \sum_p \frac{-n_F(\xi) + n_F(\xi_p + i\omega_m)}{i\omega - \xi_{p_1} - \xi_{p_2}}$$

$$= -\frac{1}{L^d} \sum_p \frac{1 - n_F(\xi_p) - n_F(\xi_{q-p})}{i\omega_n - \xi_p - \xi_{q-p}}$$

b) (from problem) "Density-Density Correlation function"

Identities for fermi:

$n_F(-x) = 1 - n_F(x)$

$n_F(x + i\omega_n) = n_F(x)$

$$\chi^d_{\omega,q} = -\frac{T}{L^d}\sum_{\omega_m, p} G_0(i\omega_m, p)\, G_0(i\omega_m + i\omega_n, p+q)$$

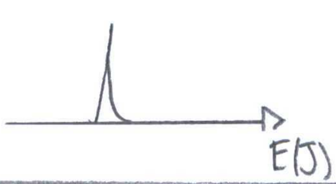
$$= -\frac{T}{L^d}\sum_{p}\sum_{\omega_m} G_0(i\omega_m, p)\, G_0(i\omega_m + i\omega_n, p+q)$$

$$= -\frac{T}{L^d}\sum_{p} \frac{n_F(\xi)}{i\omega_n + \xi - \xi_{p+q}} + \frac{n_F(\xi + i\omega_n)}{i\omega_n + \xi - \xi_{p+q}}$$

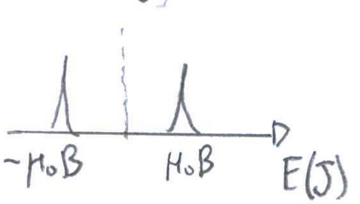
$$= -\frac{1}{L^d}\sum \frac{n_F(\xi_p) - n_F(\xi_{p+q})}{i\omega_n + \xi_p - \xi_{p+q}}$$

Problem 3.8.9 - Pauli Paramagnetism:

No magnetic field:



Magnetic field



(from problem) "Hamiltonian in magnetic field"

$$\hat{H} = -\mu_0 B_0 \hat{S}, \quad \hat{S} = \frac{1}{2} a^\dagger_{\alpha\sigma}\, \sigma_{\sigma\sigma'}\, a_{\alpha\sigma'}$$

where σ = Pauli matrices

α = orbital quantum number

μ_0 = "Bohr magneton" $= e/2m$

a) Hamiltonian:

$$\hat{H} = \hat{H}_0 + \hat{H}_Z \quad \text{.. when } \hat{H}_0 = \sum_{\alpha,\sigma} a^\dagger_{\alpha\sigma} \epsilon_\alpha a_{\alpha\sigma}$$

$$= \sum_{\alpha\sigma} a^\dagger_{\alpha\sigma} \epsilon_\alpha a_{\alpha\sigma'} - \mu_0 B \hat{S}$$

$$= \sum_{\alpha\sigma} a^\dagger_{\alpha\sigma} \epsilon a_{\alpha\sigma'} - \mu_0 B \left(\tfrac{1}{2} a_{\alpha\sigma}^\dagger \sigma_{\sigma\sigma'} a_{\alpha\sigma'} \right)$$

$$= \sum_{\alpha\sigma} a^\dagger_{\alpha\sigma} \left(\epsilon_\alpha - (\mu_0 B/2) \sigma_{\sigma\sigma'} \right) a_{\alpha\sigma}$$

Partition function:

(pg 136) "Grassmann Integrals"

$$\int d\bar{\Phi}\, d\Phi\, f(\bar{\Phi}, \Phi) = \underbrace{\det(MM')}_{} \int d\bar{v}\, dv\, f(\bar{\Phi}(v), \Phi(v))$$

"Somewhat like a u-substitution determinant"

(pg 140) "Action variable"

$$S[\Phi] = T \sum \bar{\Phi}_{an} (-i\omega_n + \xi) \Phi_{an}$$

$$= T \sum \bar{\Phi}_{an} (-i\omega_n + \xi - \mu_0 B/2) \sigma_{\sigma\sigma'} \cdot \bar{\Phi}_{an}$$

(3.76) "Partition function"

$$Z = \int D\phi \, e^{-S[\phi]}$$

$$= \int D\phi \, e^{-\int d\varphi_{an} d\varphi_{an} \, S[\psi]}$$

$$= \int D\phi \, e^{-M\int d\psi \, d\psi \, S[\psi]}$$

$$= \int D\phi \, e^{-\prod_{\alpha\sigma} \begin{pmatrix} -i\omega + \xi & \mu_0 B/2\sigma_{\sigma\sigma'} \\ \mu_0 B/2\sigma_{\sigma\sigma'} & -i\omega + \xi \end{pmatrix} \int_0^\beta \int_0^\beta d\psi \, d\psi \, \psi^* \psi}$$

$$= \int D\phi \, e^{-\beta^2 \left[(-i\omega + \xi)^2 - \left(\frac{\mu_0 B}{2}\right)^2 \right]}$$

$$= \prod_n \left(\beta^2 (-i\omega_n + \xi_n)^2 - \left(\frac{\mu_0 B}{2}\right)^2 \right)$$

similar to equation (3.76)
-unsure how to plot, or a table-

Free energy:

(3.77) "Free Energy"

$$F = -T \ln Z$$

$$= -T \ln \left[\prod_n \beta^2 (-i\omega_n + \xi)^2 - \left(\frac{\mu_0 B}{2}\right)^2 \right]$$

$$= -T \sum \ln \left[\beta^2 (-i\omega_n + \xi)^2 - \left(\frac{\mu_0 B}{2}\right)^2 \right]$$

b) Magnetic Susceptibility:

(from problem) "Total Energy change"

$\Delta E \sim -\mu_0^2 B^2 \rho(u)$

(from problem) "Magnetic Susceptibility"

$\chi \sim -\partial_B^2 \Delta E$

$= \cancel{-\mu_0} \frac{\partial^2}{\partial B^2} \Big| F_{B=0}$

$= -\frac{\partial^2}{\partial^2 B}\left[-T \sum \ln(\beta^2 [\cancel{(}-i\omega + \xi)]^2 - \frac{1}{4}(\mu_0 B)^2\right]_{B=0}$

$= -\frac{1}{2}\mu_0^2 T \sum \frac{1}{(-i\omega_n + \xi)^2}$

(Pg 142) "Isolated Singularities"

$S = \sum h(\omega_n)$

$= -\frac{1}{2}\mu_0^2 T \sum_\alpha \frac{1}{(-i\omega_n + \xi)^2}$ ooo where $h(\omega_n) = n_F(\epsilon_\alpha)$

$= -\frac{1}{2}\mu_0^2 \sum_\alpha n_F(\epsilon)$

$\lim_{T \to 0} \chi = -\frac{\mu_0^2}{2}$

Problem 3.8.10 - Boson-Fermion duality:

Matsubura:

Re(Frequency) ⟷ Im(Frequency)

Fourier:

Re(Frequency) ⟷ Re(Time)

Laplace:

Re(Time) ⟷ Im(Time)

Hankel:

Re(position) ⟷ Re(Scaled Position)

Wavelet:

Re(time) ⟷ Re(Scaled time)

"Transforms

(3.71) "Partition function"

$$Z = \int D\psi \, e^{-S[\psi]}$$

$$= \int D\psi \, e^{-\int dx d\tau \, \psi^{\dagger}(\partial_{x_0} \pm i\partial_x)\psi}$$

$$= \int D\psi \, e^{-\int dx d\tau \, \psi^{\dagger} G(0,0,t)^{-1} \psi} \quad \ldots (3.29)$$

$$G_{\pm}(x,\tau) = (\partial_\tau \pm \partial_x)^{-1}$$

(3.95) "Correlation function"

$$C(x,\tau) \equiv \langle (\psi_+^{\dagger} \psi_-)(x,\tau)(\psi_-^{\dagger}\psi_+)(0,0) \rangle_{\psi}$$

$$= G_0(x,\tau) \circ G_0(x,\tau)$$

b) (From problem) "Action in a Bosonic field"

$$S[\theta] = \frac{1}{2c} \int dx d\tau \left[(\partial_\tau \theta)^2 + (\partial_x \theta)^2 \right]$$

$$= \frac{L}{2cT} \sum_{x,\tau} \left[q^2 + w^2 \right]$$

$$= G(x,\tau)^{-1}$$

Correlation function:

$$K(x,\tau) = \langle \Theta(x,\tau)\Theta(0,0) - \Theta(0,0)\,\Theta(0,0)\rangle$$

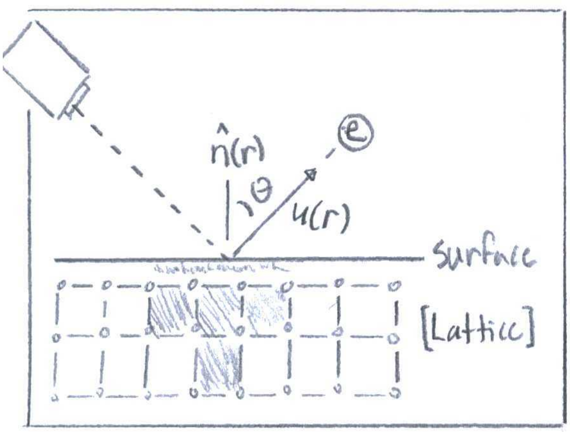
$$= G(x,\tau)$$

C) (from problem) "Correlation Function"

$$C(x,\tau) = T^2 \langle e^{2i\Theta(x,\tau)}, e^{-2i\Theta(0,0)}\rangle$$

$$= T^2 \langle e^{2i(\Theta(x,\tau) - \Theta(0,0))}\rangle$$

Problem 3.8.11 - Electron Phonon Coupling:



"Electron-phonon Coupling"

(from problem) "Electron-phonon Hamiltonian"

$$\hat{H}_{el\text{-}ph} = \gamma \int d^d r\, \underbrace{\hat{n}(r)\, \nabla\cdot u(r)}_{\text{"Angle between normal"}} = \gamma \sum_{q,j} \frac{iq_j}{(2m\omega_q)^{1/2}}\, \hat{n}_q (a_{q,j} + a^\dagger_{-q,j})$$

where $u(r) = e_j (a_{q,r} + a^\dagger_{-q,r}) / (2m\omega_q)^{1/2}$

(from problem) "Phonon Hamiltonian"

$$\hat{H}_{ph} = \sum_{q,i} \omega_q a^\dagger_q a_q$$

a) (3.71) "Partition function"

$$Z = \int D\phi \, e^{-S[\phi]}$$

$$= \int D\phi \, e^{-(S[\phi]_{el\text{-}ph} + S[\phi]_{ph} + S[el])}$$

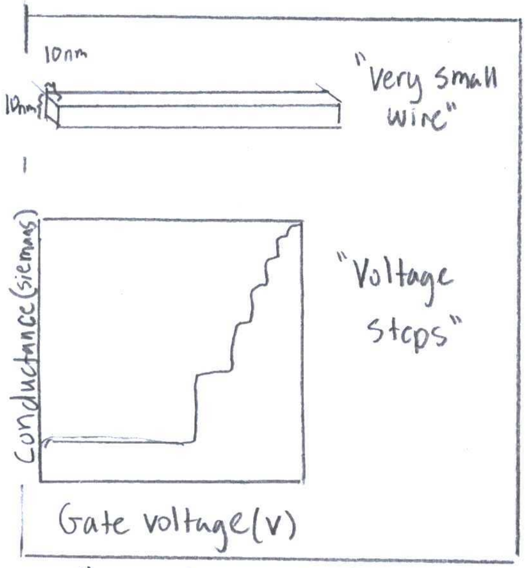
$$= \int D\phi \, e^{-\int d\phi \, \gamma \frac{iq_j}{(2mq)^{1/2}} \hat{n}_q (a_q + a^\dagger_{-q}) + w a_q^\dagger a_q + \frac{\hat{p}^2}{2m}}$$

$$= \int D\phi \, e^{-\int d\phi \, \gamma \frac{iq}{(2mq)^{1/2}} \rho_q (\phi_{q_i} + \Phi_{q_i}) + w a_q^\dagger a_q + \frac{\hat{p}^2}{2m}}$$

b) $S_{eff}[\phi] = S_{el}[\phi] + S_{ph}[\phi] + S_{el\text{-}ph}[\phi]$

$$= \frac{\hat{p}^2}{2m} + \hbar w - \frac{\gamma^2}{2mw} \sum \frac{q^2}{\quad} \hat{n} \rho \rho$$

Problem 3.8.12 - Disordered Quantum Wires:



"Quantum wire"

(3.86) "Action Potential in Fermionic gas."

$$S_0[\psi] = \int d^2x \, \psi^\dagger (\sigma_0 \partial_{x_0} - i\sigma_3 \partial_{x_1}) \psi$$

$$= \int d^2x \, \bar{\psi} (\sigma_1 \partial_{x_\alpha} - \sigma_2 \partial_{x_1}) \psi$$

(3.90) "Action with Second quantization"

$$S_{int}[\psi] = \frac{1}{2} \sum_s \int dx d\tau \, (g_5 \hat{\rho}_s \hat{\rho}_{\bar{s}} + g_4 \hat{\rho}_s \hat{\rho}_s)$$

(from problem) "Action in impurity"

$$S_{imp}[\psi] = \int d\tau \left(V_+ \psi_+^\dagger \psi_+ + V_1 \psi_+^\dagger \psi_+ \right.$$
$$\left. + V \psi_+^\dagger \psi_- + \bar{V} \psi_-^\dagger \psi \right)$$

a) If $\psi_+ = e^{-iV_+\theta(x)/V_F}$, $S_{impurity}[\phi]'$ is without a $V_+ \psi_+^\dagger \psi_+$ term. The forward scattering term is not present in a "one-dimensional" wire.

b) (pg 151) "Lagrangian form of action"

$$S[\theta] = \frac{1}{2\pi} \int d\tau dx \left((\partial_\tau \theta)^2 + (\partial_x \theta)^2 \right)$$

$$S_{TOT}[\theta] = \frac{1}{2\pi} \int dx d\tau \left(v (\partial_x \theta)^2 + (\partial_\tau \theta)^2 \right) + S_{imp}[\phi]$$

c) (from problem) "Action Integral"

$$e^{-S[\theta]} = e^{-S_{imp}[\theta]} \cdot \int D\theta dk\, e^{-\int dx d\tau \left(\frac{1}{2\pi g} (v(\partial_x\theta)^2 + v^{-1}(\partial_\tau \theta)^2) \cdots - ik(\hat{\theta} - \theta)\delta(x) \right)}$$

$$= e^{-S_{imp}[\theta]} \cdot \int Dk\, e^{i\sum k\theta}\, e^{\left(-\frac{\pi g T}{2L} \sum (vq^2 + v^{-1}\omega^2)|k_n|^2 \right)}$$